CHAPTER SIX

ELASTICITY

When an external force acts on a body which is not free to move, the molecules of the body are shifted to new positions. This change results in the deformation of the body because its size and shape changes. The applied force which caused changes in the body is referred to as a *deforming force.* Some materials become permanently deformed even when the deforming force has been removed. There are certain materials which tend to recover their size and shape after the deforming force has been removed. Such materials are called elastic materials because the molecules tend to regain their original positions. Whenever a deforming force is applied to an elastic body, an internal force or a restoring force is set up in the body which acts in opposite direction to the deforming force.

Elasticity is *the property of a material by virtue of which it regains its original length, volume and shape after the deforming forces have been removed*.

Stress

Stress is a measure of the strength of a force that is causing a deformation.

Stress is defined as *the deforming force acting per unit area of a body.* Its SI unit is Nm^{-2}.

$$Stress = \frac{Deforming\ Force}{Cross\ sectional\ Area} \qquad \textbf{6.1}$$

(a) Tensile stress

In tensile stress the deforming force stretches the body in one direction. This causes the body to increase in length. This is also known as longitudinal stress.

Consider a long wire fixed to a rigid support at one end as shown in Figure 6.1. Let a deforming force *F* be applied normally outwards at the free end.

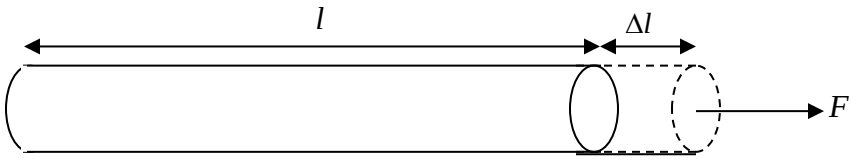


Figure 6.1

Let us also assume that the wire extends a length Δl in the direction of the force. Then,

$$Tensile\ stress = \frac{Deforming\ Force}{Cross\ sectional\ Area} \qquad \textbf{6.2}$$

(b) Compressional stress

It is the restoring force developed per unit cross sectional area of a body when the body is compressed. The deforming force causes the length of the body to decrease.

(c) Tangential or shearing stress

When the deforming force acts tangentially over an area of a body, it causes the body to get sheared through an angle θ. Such a deformation results from a tangential stress.

(d) Hydrostatic or Bulk stress

In this type of stress, the deforming force acts uniformly and equally from all sides of the body's surface. Bulk stress causes the entire body's size to decrease.

Strain

It is the fractional deformation resulting from a stress. This means that any type of stress has a corresponding strain. Strain has no unit.

(a) Tensile or longitudinal strain

When the deforming force produces a change in length, then the strain becomes the extension per unit length. Suppose that a wire of length l is stretched to a new length $l + \Delta l$ under stress. Then,

$$Tensile\ strain = \frac{change\ in\ length}{original\ length} = \frac{\Delta l}{l} \qquad \textbf{6.3}$$

Therefore, within the *elastic limit*, tensile strain can be defined as *the ratio of the change in length to the original length.*

(b) Shear strain

If the deforming force produces a change in the shape of a body, then the strain is measured by the angle through which a line originally perpendicular to the fixed plane is turned. Within the elastic limit, shear strain is measured by the ratio of the relative displacement of one plane to its distance from the fixed plane.

So, shear strain can be defined as *the relative displacement between two parallel planes which are a unit distance apart.*

(c) Bulk strain

When the deforming force produces a change in volume, then the strain becomes the bulk strain. Suppose a body of volume V is compressed to a new volume $V-\Delta V$ under stress. Then,

$$Bulk\ strain = \frac{change\ in\ volume}{original\ volume} = \frac{\Delta V}{V} \qquad \textbf{6.4}$$

Therefore, within the elastic limit, bulk strain can be defined as *the ratio of the change in volume to the original volume.*

Example 6.1. What are stress and strain?

Solution

Stress is a measure of the strength of the agent that is causing a deformation. If a force F is applied, possibly obliquely, to a surface of area A, then

$$\text{stress} = \frac{\text{force}}{\text{area of surface on which force acts}} = \frac{F}{A}$$

The units of stress are those of pressure.

Strain is the fractional deformation resulting from a stress. It is measured by the ration of the change in some dimension of a body to the original dimension in which the change occurred.

$$\text{strain} = \frac{\text{change in dimension}}{\text{original dimension}}$$

Strain has no units because it is a ratio of like quantities.

PLASTICITY

Plasticity is *the property of a body remaining deformed even after the deforming force has been removed.*

Plastic bodies do not show any tendency to recover their original shape and size when the force acting on them is removed. Forces exerted on plastic materials cause dislocation in their crystal lattices. If a metal is deformed beyond its elastic limit, it exhibits plastic behavior.

Hooke's law

The law states that *the extension is proportional to the force or tension in a wire if the elastic limit is not exceeded* OR *provided the elastic limit is not exceeded, stress is directly proportional to strain.*

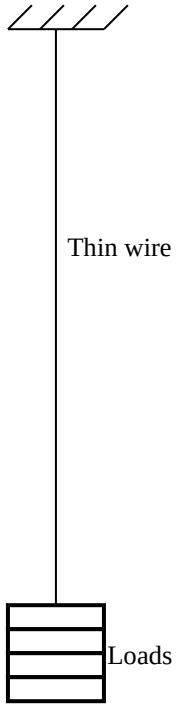


Figure 6.2

Let us consider a wire supported at a fixed point as shown in Figure 6.2. When a load is placed on the wire, it extends and the extension ***e*** is read as change in length. Figure 6.3 shows a typical Stress – Strain graph if the thin wire is loaded with increasing weights. From the graph, within region OA, the wire obeys Hooke's law. However a further increase in load increases the tension in the wire until a point is reached where the proportionality limit is exceeded i.e., the wire cannot return to its initial state or position when it is off-loaded. This is the *yield point* which is indicated by kinks in the wire. A further increase in the load causes *plastic deformation* where the wire stretches and increases in length permanently and becomes thinner. Finally the wire breaks after reaching the *breaking point* with more loads on the wire. The break is due to the inability of the tension in the wire to support the applied load on the wire. The tensile stress in the wire is the force per unit area. Tensile strain is extension per unit length. The modulus of elasticity is a measure of the stiffness of the material**.** From the graph, three important mechanical properties can be obtained. (**i**) yield strength (**ii**) tensile strength (**iii**) modulus of elasticity.

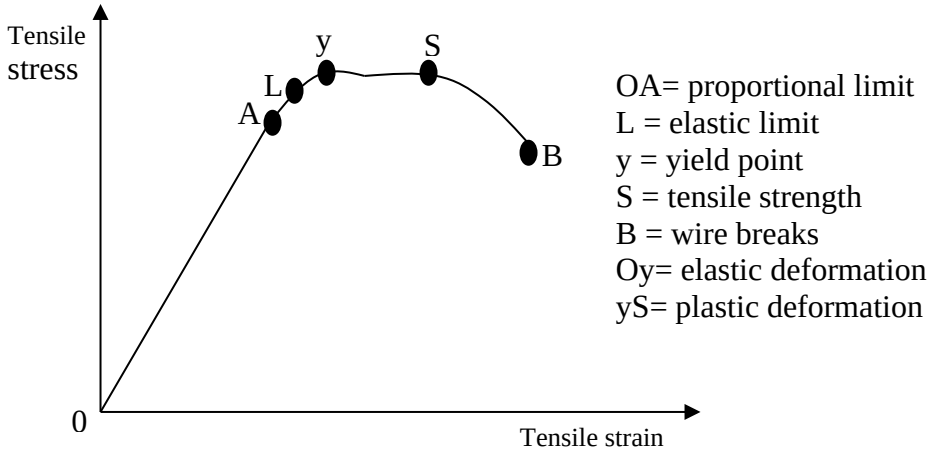


Figure 6.3 A typical Stress verses Strain graph

Example 6.2. Define elasticity, Hooke's law, elastic limit, and ultimate strength.

Solution

Elasticity is that property by virtue of which a body returns to its original size and shape when the forces that deformed it are removed.

Hooke's law can be stated in terms of stress and strain. If the system obeys Hooke's law, then stress $\propto$ strain. We then define a constant the modulus of elasticity, by the relation

$$\text{modulus of elasticity} = \frac{\text{stress}}{\text{strain}}$$

The modulus has the same units as stress. In the case of pure tension, the modulus is called Young's modulus.

The elastic limit is the smallest value of the stress required to produce a permanent distortion in a body. When a stress in excess of this limit is applied, the body will not return exactly to its original state after the stress is removed.

The ultimate strength is the smallest value of stress required to break the body.

Young's Modulus of elasticity

From Hooke's law, within the elastic limit stress is proportional to strain.

That is, Force $\mathbf{F} \propto$ extension $\mathbf{e}$.

$$\mathbf{F} = k\mathbf{e} \qquad \textbf{6.5}$$

Dividing both sides of Equation 6.5 by the area A

$$\frac{\mathbf{F}}{A} = \frac{k}{A}\mathbf{e}$$

but the area $A = l \times l$

$$\frac{F}{A} = \frac{k}{l}\frac{e}{l}$$

$$stress = \frac{F}{A}, \quad \frac{k}{l} = constant, K, \quad strain = \frac{e}{l}$$

Therefore, $stress = K \times strain$

$$\text{Constant } K = \frac{Stress}{Strain} \qquad \textbf{6.6}$$

K is a constant known as *Modulus of Elasticity*.

If the change is an increase in length, then *K* becomes *Young Modulus of Elasticity* denoted by Y .

Therefore, Young's Modulus of Elasticity is defined as

$$Y = \frac{Tensile\,Stress}{Tensile\,Strain} \qquad \textbf{6.7}$$

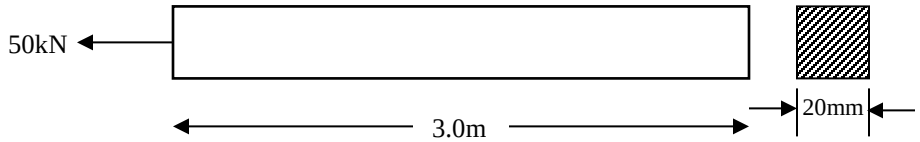
Mathematically,

$$Y = \frac{F}{A} \div \frac{\Delta l}{l} = \frac{F}{A} \cdot \frac{l}{\Delta l} \qquad \textbf{6.8}$$

Note that Young's Modulus applies to only solids.

Example 6.3. A prismatic bar with square cross section (20mm x 20mm) and length L= 3.0m is subjected to an axial tensile force of 50kN. The measured elongation of the bar is ΔL= 1.5mm. Calculate the tensile stress and strain in the bar. Hence find the Young's modulus of the prismatic bar.

Solution



$$Tensile\,stress = \frac{Force}{cross\,\text{sectional}\,Area} = \frac{50kN}{(20\text{ x }20)mm^2} = 125MPa$$

$$Tensile\,strain = \frac{elongation}{originallength} = \frac{1.5x10^{-3}m}{3.0m} = 5.0x10^{-4}$$

$$Young's\,modulus = \frac{Tensile\,stress}{Tensile\,strain} = \frac{125MPa}{5.0x10^{-4}} = 25MPa$$

Example 6.4. What is Young's modulus?

Solution

Young's modulus (or tensile modulus) describes the length elasticity of a material. Suppose a wire or rod of original length L and cross sectional area A elongates an amount ΔL under a stretching force F applied to its end. Then

$$\text{tensile stress} = \frac{F}{A} \qquad \text{tensile strain} = \frac{\Delta L}{L}$$

and $$\text{Young's modulus} = Y = \frac{\text{stress}}{\text{strain}} = \frac{F/A}{\Delta L/L} = \frac{FL}{A\Delta L}$$

The value of Y depends only on the material of the wire or rod not on its dimensions.

Example 6.5. An iron rod 4 m long and 0.5 cm^2 in cross section stretches 1 mm when a mass of 225 kg is hung from its lower end. Compute Young's modulus for the iron.

Solution

$Y = \frac{\text{stress}}{\text{strain}} = \frac{F/A}{\Delta L/L}$ where $\frac{\Delta L}{L} = (1 \times 10^{-3}\text{m})/4\text{ m} = 0.25 \times 10^{-3}$, $A = 0.5 \times 10^{-4}\text{m}^2$, and $F = (225\text{kg})(9.8\,\text{m/s}^2) = 2205$ N. Substituting, we get $Y = 1.76 \times 10^{11}\ \text{N/m}^2 = 176$ MPa

Example 6.6. If we consider the rod of example **Example 6.5** to be a spring, within its elastic limit, what is the effective force constant of this spring?

Solution

$F = YA\Delta L/L = k\Delta L$, where k is by definition the force constant. Thus

$$k = \frac{YA}{L} = \frac{(1.76 \times 10^{11}\ \text{N/m}^2)(0.5 \times 10^{-4}\text{m}^2)}{4\text{ m}} = 2.2\ \text{MN/m}$$

Shear Modulus of Elasticity or Modulus of Rigidity

If the deforming force produces a change in the shape of the body, then from Equation 6.6, the constant *K* becomes *Shear Modulus of Elasticity,* which is denoted by G.

Therefore Equation 6.6 becomes,

$$G = \frac{ShearStress}{ShearStrain} \qquad \textbf{6.9}$$

Let us consider a rectangular block with the front vertical face labeled QRST as shown in Figure 6.4a. If a tangential force **F** is applied on the upper parts, say ST, of the block, a reaction force will

also be set up at the bottom face, QR. These two tangential forces will cause the shape of the block to change but the volume will be unaltered.

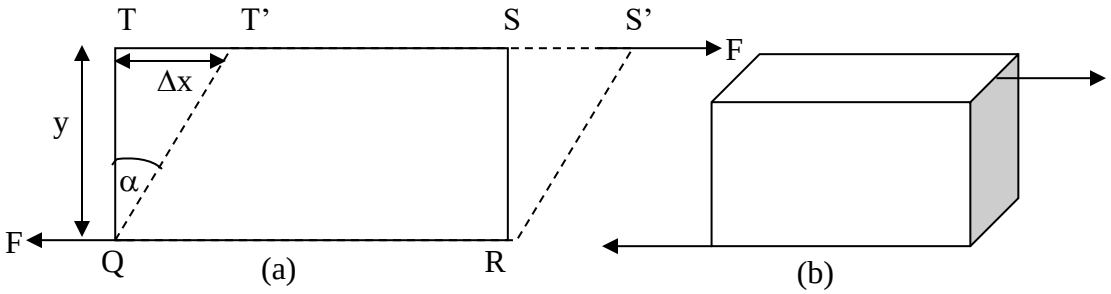


Figure 6.4 Shear deformation of a rectangular block by a tangential force

As you go up from the bottom each layer slides past the neighbor by the same margin making the block to suffer a tangential shear.

Therefore, the shear stress is given by the tangential force per unit area as F/A.

Also, the corresponding shear strain is given by the tangent of the angle of shear as $\tan\alpha = \Delta x/y$

where α is the angular displacement from the initial position.

$$G = \frac{Shear\ Stress}{Shear\ Strain} = \frac{F}{A} \div \frac{\Delta x}{y} = \frac{F}{A} \cdot \frac{y}{\Delta x} \qquad \textbf{6.10}$$

So, the Shear Modulus or Modulus of rigidity is defined *as the ratio of the shear stress to the shear strain.*

Note that Shear Modulus applies to solids and liquids.

Bulk Modulus

Bulk Modulus applies to a change in volume of a body if a deforming force acts uniformly over the surface of the body. Suppose we apply a force normally and uniformly over an area of a sphere of volume V such that the volume decreases by ΔV. In this case there will be no change in shape of the sphere. Such a uniform pressure acting normally to the entire surface of the body is termed *Hydrostatic pressure*.

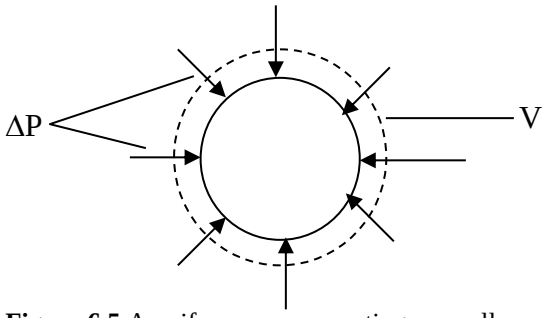


Figure 6.5 A uniform pressure acting normally to the entire surface of the body

Considering the volume change, the constant K in Equation 6.6 becomes the Bulk Modulus of elasticity.

Therefore, Bulk Modulus of elasticity is,

$$K = \frac{Bulk\ Stress}{Bulk\ Strain}. \qquad \textbf{6.11}$$

The pressure applied to compress the body is called the Bulk Stress which is given by $\Delta P = {}^{F}\!/_{A}$.

The corresponding fractional change in volume is called the Bulk strain and it is given by $\left(-\frac{\Delta V}{V}\right)$

. The negative sign denotes the decrease in volume.

$$K = \frac{Bulk\ Stress}{Bulk\ Strain} = \frac{F}{A} \div \frac{\Delta V}{V} = \frac{F}{A} \cdot \frac{V}{\Delta V} = \frac{\Delta P}{{}^{\Delta V}\!/_{V}} \qquad \textbf{6.12}$$

So, the Bulk Modulus is defined as *the ratio of normal stress to the volumetric strain.*

Note that the Bulk Modulus applies to all phases of matter.

Example 6.7. Determine the fractional change in volume as the pressure of the atmosphere (0.1 MPa) around a metal block is reduced to zero by placing the block in vacuum. The bulk modulus for the metal is 125000 MPa.

Solution

$B = \frac{-\Delta p}{\Delta V/V}$ or $\frac{\Delta V}{V} = -\frac{\Delta p}{B} = \frac{-(-0.1)}{125000} = 8 \times 10^{-7}$

Example 6.8. How large a pressure must be applied to water if it is to be compressed by 0.1 percent? What is the ratio of this pressure to atmospheric pressure, 101 kPa? The bulk modulus of water is 2100 MPa.

Solution

The volume strain $\frac{\Delta V}{V} = 1.0 \times 10^{-3}$; set $B = p/(\Delta V/V_0)$ to find

$p = (2.1 \times 10^9)(1.0 \times 10^{-3}) = 2100$ kPa.

Dividing by the atmospheric pressure, the ratio is 21.

Exercise 6

1. A metal wire 75cm long and 0.13cm in diameter stretches 0.035cm when a load of 8.0kg is hung on its end. Find the stress, the strain, and the Young's modulus (Y) for the material of the wire. **(Ans: stress=5.91 $\times 10^7$Pa; strain=4.67$\times 10^{-4}$; y=127GPa)**

2. A solid cylindrical steel column is 4.0m long and 9.0cm in diameter. What will be its decrease in length when carrying a load of 80,000kg? (Y=1.9$\times 10^{11}$Pa) **(Ans: Δ L=2.6mm)**

3. A 60Kg woman stands on a light, cubical box that is 5cm on each edge. The box sits on the floor. What pressure does the box exerts on the floor? **Ans: P=240kPa)**

4. An iron rod 4m long and 0.5cm^2 in cross section stretches 1mm when a mass of 225kg is hung from its lower end. Compute Young's modulus for the iron. **Ans: 176GPa)**

5. Find the density and specific gravity of ethyl alcohol if 63.3g occupies 80.0mL. **(Ans: 791kgm^{-3}; 0.791)**

6. A load of 50kg is applied to the lower end of a steel rod 80cm long and 0.60cm in diameter. How much will the rod stretch? Y=190GPa for steel. **(Ans: 73µm)**

7. Compute the volume change of a solid copper cube, 40mm on each edge, when subjected to a pressure of 20MPa. The bulk modulus for copper is 125GPa. **Ans: -10mm^3)**

6.10 The compressibility of water is 5×10^{-10}m^2/N. find the decrease I volume of 100mL of water when subjected to a pressure of 15MPa.
(Ans: 0.75mL)

6.11 A body of mass 2kg is attached to the end of a vertical wire of length 2m and diameter 0.64mm and the extension is 0.60mm. Calculate (a) tensile stress (b) tensile strain (c) Young Modulus (E) [Take g = $9.8ms^{-2}$]

6.12 A vertical wire is suspended from one end . a weight of 20N is attached to the other end to stretch the wire. If the weight extends the wire by 1mm, calculate the energy gained by the wire.

6.13. If a force of 100N stretches a string by 0.1cm, find (a) the elastic constant (b) the work done in stretching the string 0.3cm if the elastic limit is not exceeded.

6.14 A uniform steel wire has cross sectional area $3mm^2$ and length 10m. A mass of 10kg is attached at one end. If the Young Modulus (E) of the steel is $10^{-2}Nm^{-2}$, calculate the extension produced. [Take g = $9.8ms^{-2}$]